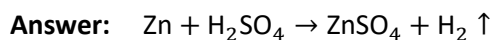


ACIDS BASES AND SALTS

SHORT QUESTIONS (1 or 2 MARKS)

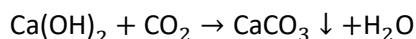
1. Write the chemical equation when zinc reacts with dilute sulfuric acid. Name the gas evolved and the test to detect it.



Gas evolved: Hydrogen; Test: It burns with a 'pop' sound.

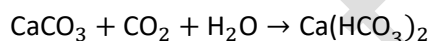
2. What happens when carbon dioxide is passed through lime water? Write the chemical equations for changes.

Answer: Short duration:



Milkiness appears.

Long duration (excess CO_2):



Milkiness disappears.

3. Define Ph. How does it indicate acidity or basicity? Give examples.

Answer: Ph is a measure of hydrogen ion concentration. $\text{Ph} < 7$ acidic (lemon juice), $\text{Ph} = 7$ neutral (pure water), $\text{Ph} > 7$ basic (soap solution).

4. Differentiate between strong and weak acids with examples.

Answer: Strong acids fully ionize (HCl), weak acids partially ionize (CH_3COOH).

5. Describe the preparation, chemical formula, and uses of washing soda.

Answer: Washing soda: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$. Prepared by carbonation of sodium hydroxide and sodium chloride. Used in glass, soap manufacture, water softening.

6. Explain the preparation and uses of bleaching powder.

Answer: Prepared by passing chlorine over dry slaked lime. Uses: disinfectant, bleaching agent in paper, textile industries.

7. What is water of crystallization? Give an example.

Answer: Water molecules incorporated in crystal lattice. Example: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (blue vitriol).

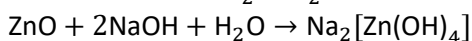
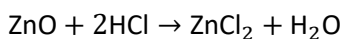
8. Write chemical equation for reaction between hydrochloric acid and sodium hydroxide. Name the salt formed.



Salt: Sodium chloride.

9. What is an amphoteric oxide? Give an example with reaction.

Answer: Oxides reacting with acid and base. Example: ZnO .



10. How does acid rain affect environment? How to reduce it?

Answer: Acid rain damages plants, aquatic life, and buildings. Control SO₂ and NO emissions from factories and vehicles.

11. Write reaction of acid with metal carbonate and metal hydrogen carbonate.

Answer: $2\text{HCl} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$

12. Observations when acid reacts with base in neutralization.

Answer: Acid and base neutralize forming salt and water; Ph becomes neutral.

13. What are indicators? Name three and colour changes in acid/base.

Answer: Litmus (red in acid, blue in base), Phenolphthalein (colourless in acid, pink in base), Methyl orange (red in acid, yellow in base).

14. Write chemical equations for reaction of acids with metals.

Answer: $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2 \uparrow$

15. Uses of baking soda.

Answer: Antacid, baking, fire extinguisher.

16. How does plaster of Paris form from gypsum? Write equation.

Answer: $2\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{\text{heat}} 2\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + 3\text{H}_2\text{O}$

17. Write ionic equation of neutralization between HCl and NaOH.

Answer: $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$

18. Give example and equation of acid reacting with metal.

Answer: $\text{Mg} + 2\text{HCl} \rightarrow \text{MgCl}_2 + \text{H}_2 \uparrow$

19. Match acids with their sources: lactic, acetic, citric, oxalic acid.

Answer: Lactic (milk), Acetic (vinegar), Citric (citrus fruits), Oxalic (spinach).

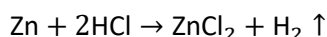
20. Why is HCl called a strong acid and acetic acid weak?

Answer: HCl ionizes completely; acetic acid partially ionizes in water.

2 MARKS

1. Explain the reaction of zinc with dilute hydrochloric acid. Write the chemical equation and the test of the gas evolved.

Answer: Zinc reacts with HCl to form zinc chloride and hydrogen gas.



Hydrogen gas burns with a 'pop' sound.

2. What is pH? Explain why pH is important in everyday life. Give two examples.

Answer: pH indicates acidity/basicity on a scale of 0 to 14.

Important because it affects soil fertility, digestion, etc.

Examples: Lemon juice (acidic), soap (basic).

3. Describe how bleaching powder is prepared and list three of its uses.

Answer: Prepared by passing chlorine gas over dry slaked lime.

Uses: Disinfectant, bleaching agent, water treatment.

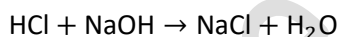
4. Write the chemical equation when carbon dioxide reacts with lime water. Explain the changes occurring.

Answer: $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 \downarrow + \text{H}_2\text{O}$

Milkiness appears. On excess CO_2 , calcium bicarbonate forms, clearing the solution.

5. Explain the concept of neutralization with a balanced chemical equation.

Answer: When acid reacts with base, salt and water form.

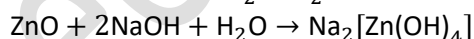


6. What are amphoteric oxides? Give two examples and their reactions with acids and bases.

Answer: Amphoteric oxides react with both acids and bases.

Example: ZnO , Al_2O_3 .

Reactions:



7. What is water of crystallization? Explain its importance with an example.

Answer: Water molecules in crystal lattice.

Example: $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

Importance: Helps identify purity, and some salts lose water on heating.

8. Give the reaction of an acid with a metal carbonate and explain the gas evolved.

Answer: $2\text{HCl} + \text{Na}_2\text{CO}_3 \rightarrow 2\text{NaCl} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$

CO_2 gas is colourless, odourless, and turns lime water milky.

9. Write the balanced equation for the preparation of plaster of Paris from gypsum.

Answer: $2\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{\Delta} 2\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O} + 3\text{H}_2\text{O}$

10. Explain the difference between strong and weak acids with examples and their dissociation in water.

Answer: Strong acids fully ionize e.g. HCl .

Weak acids partially ionize e.g. CH_3COOH .

11. How is baking soda used in daily life? Write its chemical formula.

Answer: Formula: NaHCO_3 .

Uses: Antacid, cooking, fire extinguisher.

12. What is acid rain? How does it affect plants and aquatic life?

Answer: Acid rain has low pH due to SO_2 , NO_x gases.

It harms plant leaves and aquatic animals by lowering water pH.

13. Write the reaction between hydrochloric acid and sodium hydroxide using ionic equations.

Answer: $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$

14. Describe the test and observation for detecting hydrogen gas.

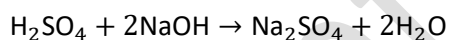
Answer: Collect gas, bring a burning splinter near it, it produces a 'pop' sound.

15. Explain the use of litmus as an indicator with examples.

Answer: Red litmus turns blue in base; blue litmus turns red in acid.

16. What happens chemically when an acid neutralizes a base? Write a reaction to illustrate.

Answer: Acid-base neutralization forms salt and water.



17. State two differences between acids and bases.

Answer: Acids taste sour, bases taste bitter.

Acids turn blue litmus red, bases turn red litmus blue.

18. What is the use of washing soda? Write its chemical formula.

Answer: Formula: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$.

Uses: Water softener, glass and soap manufacture.

19. Identify the acids as lactic, acetic, citric, oxalic with their sources.

Milk ; Vinegar ; Spinach ; Citrus fruits

Answer: Lactic - Milk

Acetic - Vinegar

Citric - Citrus fruits

Oxalic - Spinach

20. Explain why hydrochloric acid is called a strong acid while acetic acid is weak.

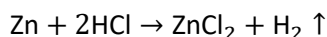
Answer: HCl ionizes completely; acetic acid ionizes partially in water.

5 Marks Question

1. Explain the reaction of acids with metals with examples and write the general chemical equation.

Answer: Acids react with metals to produce a salt and hydrogen gas.

Example:



The general equation:



2. Describe the preparation, properties, and uses of washing soda.

Answer: Washing soda is sodium carbonate decahydrate $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$.

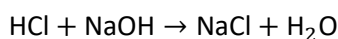
Preparation: By passing carbon dioxide through sodium hydroxide and sodium chloride solution.

Properties: It is a crystalline solid with a soapy touch, soluble in water, exhibits basic nature.

Uses: Water softener, in glass, soap, and paper industries.

3. Explain neutralization reaction with equation and importance in daily life.

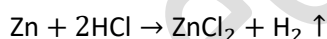
Answer: Neutralization is when acid reacts with base to produce salt and water.



Importance: Used in treating indigestion (antacids), controlling soil pH, and manufacturing salts.

4. Describe chemical properties of acids with balanced equations and observations.

Answer: Reaction with metals:



Effervescence observed.

- Reaction with metal carbonates:



Gas evolution and fizzing.

- Reaction with bases (neutralization).

5. Explain pH scale with diagram and write importance in everyday life.

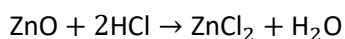
Answer: pH scale: Range 0 to 14, <7 acidic, =7 neutral, >7 basic.

Diagram shows scale with examples.

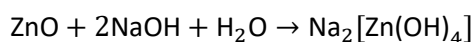
Importance: Soil pH affects farming, human blood pH maintains health, digestive system acids aid food breakdown.

6. Explain amphoteric nature of zinc oxide with reactions.

Answer: Reacts with acid:



Reacts with base:



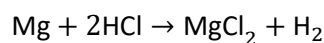
7. What is water of crystallization? Explain with example and significance.

Answer: Water molecules bound in crystalline structure, e.g., $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$.

Significance: Determines crystal properties, loss of water changes color and texture.

8. Write balanced chemical equations for reaction of acids with metals, metal carbonates, and bases.

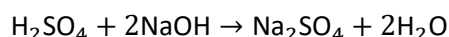
Answer: Acid + metal:



- Acid + metal carbonate:

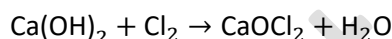


- Acid + base:



9. Describe the preparation and uses of bleaching powder with chemical equations.

Answer: Prepared by passing chlorine gas over slaked lime:



Uses: Disinfectant, bleaching agent in paper/cloth, water purification.

10. Explain how plaster of Paris is prepared from gypsum and its uses.

Answer: Heating gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) gives plaster of Paris ($\text{CaSO}_4 \cdot \frac{1}{2}\text{H}_2\text{O}$).



Uses: Making casts for fractures, ceiling decorations, Molds.

Case Based Questions (4 Marks)

1. Case: A student adds a few drops of blue litmus solution to a liquid and observes that it turns red.

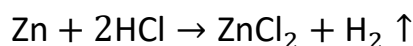
Question: What does this observation indicate about the liquid? What will happen if red litmus is added?

Solution: The liquid is acidic as blue litmus turns red in acid. Red litmus will remain red in this liquid.

2. Case: When dilute hydrochloric acid is added to a certain metal, effervescence is observed, and a gas is evolved that burns with a pop sound.

Question: Name the gas evolved. Write a chemical equation for the reaction if the metal is zinc.

Solution: Gas evolved is hydrogen.

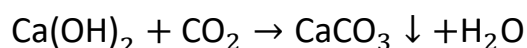


3. Case: Carbon dioxide gas is passed through lime water continuously. Initially, the lime water turns milky but later becomes clear.

Question: Explain the chemical reason for these observations with equations.

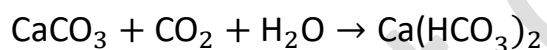
Solution:

First:



Milkiness due to insoluble CaCO_3 .

Later:

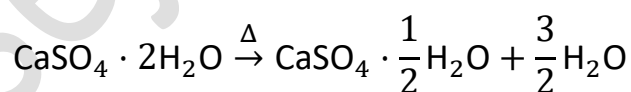


Milkiness disappears due to soluble calcium bicarbonate formation.

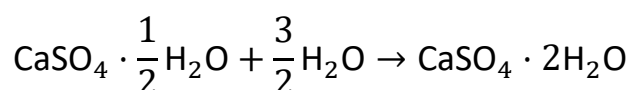
4. Case: A white powder is heated and loses water of crystallization to form a dry powder that can be powdered easily. When this dry powder is added to water, it forms a hardened solid.

Question: Identify the substance and write the chemical equations involved.

Solution: Substance is gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$). Heating forms plaster of Paris (POP).



On adding water:



5. Case: A solution turns phenolphthalein indicator pink but does not change methyl orange.

Question: What is the nature of this solution? What would be its approximate

pH range?

Solution: The solution is basic since phenolphthalein turns pink in base. Methyl orange remains yellow in base. $\text{pH} > 7$, typically 8-10.

6. Case: A student adds baking soda to lemon juice and observes effervescence. Question: What gas is evolved? Write the chemical equation and identify the type of reaction.

Solution: Gas evolved is carbon dioxide (CO_2).

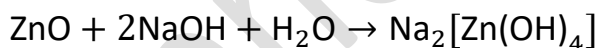


Reaction is acid-base neutralization.

7. Case: A solution of sodium hydroxide is added to a metal oxide and the metal oxide dissolves forming a solution.

Question: Identify the type of metal oxide. Write an example with equations.

Solution: The metal oxide is amphoteric. Example: Zinc oxide.



8. Case: A farmer notices the leaves of plants turning yellow while the soil pH is found to be very low.

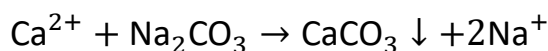
Question: What is the problem? Suggest a solution.

Solution: Soil is acidic (low pH), causing nutrient deficiency. Solution: Add lime (CaO) to neutralize acidity.

9. Case: A person uses washing soda to remove permanent hardness from water.

Question: Explain the chemical reason behind this and the reaction involved.

Solution: Washing soda reacts with bicarbonates and chlorides of calcium and magnesium to remove hardness as insoluble carbonates.



10. Case: A chemical spills on the skin causing burning and irritation. The chemical was found to be able to turn red litmus blue.

Question: Identify the chemical type and what immediate first aid should be given.

Solution: Chemical is a base (alkali). First aid: Wash the affected area immediately with plenty of water to dilute and remove it.

Multiple choice questions (1 marks)

1. Which acid is present in tomatoes?
(A) Citric Acid (B) Oxalic Acid (C) Lactic Acid (D) HCl
Answer: (B) Oxalic Acid
2. Which of the following is a strong acid?
(A) HCl (pH 1) (B) CH_3COOH (pH 5) (C) Lemon juice (pH 2.2) (D) Pure milk (pH 6)
Answer: (A) HCl
3. $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ is known as –
(A) Baking soda (B) Baking powder (C) Washing soda (D) Bleaching powder
Answer: (C) Washing soda
4. pH value less than 7 indicates that the solution is –
(A) Neutral (B) Acidic (C) Basic (D) None of these
Answer: (B) Acidic
5. Which one of the following will turn red litmus blue?
(A) Vinegar (B) Baking soda solution (C) Lemon juice (D) Soft drinks
Answer: (B) Baking soda solution
6. Which acid is used in the treatment of bone marrow and scurvy diseases?
(A) Citric acid (B) Hydrochloric acid (C) Lactic acid (D) Tartaric acid
Answer: (A) Citric acid
7. Which one of these is an organic acid?
(A) Citric acid (B) Hydrochloric acid (C) Lactic acid (D) Both (A) and (C)
Answer: (D) Both (A) and (C)

8. Which salt does not contain water of crystallization?
(A) Blue vitriol (B) Baking soda (C) Washing soda (D) Gypsum
Answer: (B) Baking soda
9. Which one of the following is a base?
(A) HCl (B) NaOH (C) CH_3COOH (D) CO_2
Answer: (B) NaOH
10. Which gas is evolved when an acid reacts with a metal?
(A) Oxygen (B) Carbon dioxide (C) Hydrogen (D) Nitrogen
Answer: (C) Hydrogen
11. What is the product when baking soda is heated?
(A) $\text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$ (B) $\text{NaOH} + \text{H}_2$ (C) $\text{NaCl} + \text{CO}_2 + \text{H}_2\text{O}$ (D) $\text{NaHCO}_3 + \text{O}_2$
Answer: (A) $\text{Na}_2\text{CO}_3 + \text{CO}_2 + \text{H}_2\text{O}$
12. Which indicator turns red in acidic medium?
(A) Phenolphthalein (B) Litmus (C) Methyl orange (D) Both (B) and (C)
Answer: (D) Both (B) and (C)
13. Which of the following is used as an antacid?
(A) NaOH (B) NaHCO_3 (C) HCl (D) H_2SO_4
Answer: (B) NaHCO_3
14. The acid present in vinegar is –
(A) Acetic acid (B) Citric acid (C) Lactic acid (D) Oxalic acid
Answer: (A) Acetic acid
15. Washing soda is used –
(A) To soften water (B) To cure indigestion (C) In bleaching (D) As an antiseptic
Answer: (A) To soften water
16. Plaster of Paris is obtained by heating –
(A) Gypsum (B) Baking soda (C) Bleaching powder (D) Washing soda
Answer: (A) Gypsum
17. Which of the following turns blue litmus red?
(A) Acid (B) Base (C) Salt (D) Water
Answer: (A) Acid

18.Salts are formed by the reaction of –

(A) Acid and Acid (B) Base and Base (C) Acid and Base (D) None of these

Answer: (C) Acid and Base

19.Which salt is used in baking?

(A) Baking soda (B) Washing soda (C) Bleaching powder (D) Plaster of Paris

Answer: (A) Baking soda

20.Which salt contains sodium carbonate and water molecules?

(A) Baking soda (B) Washing soda (C) Bleaching powder (D) Sodium chloride

Answer: (B) Washing soda

Assertion And Reason (1 Marks)

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true R is false.

(d) A is false R is true.

1.

Assertion (A): Acid turns blue litmus red.

Reason (R): Acids release H^+ ions in aqueous solution.

Answer: Both A and R are true and R is the correct explanation of A.

2.

Assertion (A): Washing soda is sodium bicarbonate.

Reason (R): Washing soda contains 10 molecules of water of crystallization.

Answer: A is false, R is true. (Washing soda is sodium carbonate decahydrate, $Na_2CO_3 \cdot 10H_2O$)

3.

Assertion (A): Hydrochloric acid reacts with zinc to produce hydrogen gas.

Reason (R): Metals above hydrogen in reactivity series displace hydrogen from

acids.

Answer: Both A and R are true and R is the correct explanation of A.

4.

Assertion (A): Phenolphthalein remains colourless in acidic and basic solution.

Reason (R): Phenolphthalein is a universal indicator.

Answer: Both A and R are false.

5.

Assertion (A): Bleaching powder is used in water treatment.

Reason (R): Bleaching powder kills germs and makes water safe for drinking.

Answer: Both A and R are true and R is the correct explanation of A.

6.

Assertion (A): Acids taste bitter.

Reason (R): Acids contain H^+ ions.

Answer: A is false, R is true. (Acids taste sour, bases taste bitter)

7.

Assertion (A): Lime water turns milky when passed with CO_2 gas.

Reason (R): $Ca(OH)_2$ reacts with CO_2 to form soluble calcium bicarbonate.

Answer: A is true, R is false. (Milky is due to insoluble calcium carbonate, $CaCO_3$)

8.

Assertion (A): Baking soda is used as an antacid.

Reason (R): Baking soda neutralizes excess stomach acid.

Answer: Both A and R are true and R is the correct explanation of A.

9.

Assertion (A): Water of crystallization is always removable by heating.

Reason (R): Heating removes water molecules loosely held in crystals.

Answer: Both A and R are true and R is the correct explanation of A.

10.

Assertion (A): pH of pure water is 7.

Reason (R): Pure water is neutral with equal concentrations of H^+ and OH^- ions.

Answer: Both A and R are true and R is the correct explanation of A.

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